MAHMOUD MOSTAFA TAYEE

AI Engineer | Machine Learning | AI Systems Engineering

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PROFESSIONAL SUMMARY

AI Engineer specializing in building and optimizing **production-grade AI/ML systems**, with expertise in **deep learning, computer vision, real-time inference, and edge/cloud deployment**. Experienced in scalable AI applications, performance optimization, multi-camera analytics, and NVIDIA DeepStream. Currently pursuing a **Master's in Informatics (Artificial Intelligence)** at Nile University, with research focused on real-time data processing and scalable AI systems. Brings additional experience in **C/C++, Python, embedded systems, and CI/CD**, combining AI development with systems-level engineering.

EXPERIENCE

Senior AI Engineer

URCA AI

July 2026 – Present | Newark, Delaware, USA | Remote

- **Led the end-to-end development of real-time computer vision systems** for intelligent surveillance and traffic analytics, spanning AI model integration, video processing architecture, inference optimization, and production deployment.
- Architected and implemented **GPU-accelerated multi-camera video analytics pipelines** using NVIDIA DeepStream and GStreamer, processing live RTSP/IP camera streams for real-time object detection, classification, multi-object tracking, and cross-camera analytics.
- Researched, evaluated, and integrated **state-of-the-art computer vision and Vision-Language Models (VLMs)** for visual scene understanding, contextual event interpretation, and multimodal video analytics.
- Developed and evaluated **domain-specific model adaptation and fine-tuning workflows**, leveraging pretrained architectures, task-specific datasets, and parameter-efficient fine-tuning techniques to improve performance on surveillance use cases.
- Designed and implemented **model optimization and deployment workflows** using ONNX and NVIDIA TensorRT, exploring reduced-precision inference, GPU resource optimization, and throughput-latency trade-offs for efficient deployment.
- Established **systematic AI evaluation and benchmarking pipelines** to assess model accuracy, detection and tracking performance, inference latency, throughput, GPU memory consumption, and overall system efficiency.
- Led the integration of **production-grade AI perception systems** into scalable video analytics platforms, addressing real-world challenges involving concurrent streams, GPU utilization, inference reliability, and operational constraints.
- Designed and delivered **technical demonstrations and proof-of-concept systems**, collaborating directly with potential customers and stakeholders to translate operational requirements into practical AI solutions.

AI Software Engineer

Ottonomi AI

November 2025 – June 2026 | Ontario, Canada | Remote

- Designed and implemented **real-time computer vision components** for traffic scene analysis and surveillance, supporting low-latency object detection, tracking, and classification
- Contributed to the design of **multi-camera analytics pipelines** for large-scale environments, supporting cross-camera tracking and spatiotemporal scene understanding
- Built and optimized deep learning models for **robust perception in real-world conditions**, including fine-tuning, data curation, and performance calibration
- Implemented **efficient inference pipelines** for real-time processing across edge and cloud environments

- Developed **event-driven data processing workflows** for scalable video stream analysis and real-time decision-making
- Conducted systematic **model evaluation and validation** to assess reliability, performance, and generalization in dynamic environments

Software Development Engineer

» **Thirdwayv Inc.**

May 2023 – November 2025 | Sheraton Al Matar, El Nozha, Cairo Governorate

- Contributed to the design and implementation of a **secure 3-tier embedded IoT architecture**, spanning a cryptographic bootloader, USB-based FOTA updater, and BLE-enabled application layer
- Built **multi-threaded C++ CLI tools** for device interfacing, firmware flashing, diagnostics, and low-level hardware interaction
- Developed **multi-threaded Python GUI and CLI tools** using Qt, Tkinter, and CustomTkinter to automate testing and validation workflows
- Designed and maintained **unit tests using Ceedling** to ensure correctness and reliability of embedded software modules
- Contributed to **CI/CD pipeline automation** by scripting build, test, and validation workflows in Python
- Performed **system-level debugging and root cause analysis** using logic analyzers, USB sniffers, and FTDI tools across hardware-software boundaries

Research and Teaching Assistant

» **Nile University | School of ITCS**

Feb 2021 – May 2023 | 6th of October City

- Actively contributed to AI and machine learning research projects under faculty supervision, focusing on **computer vision, deep learning, and real-time data processing**
- Delivered lectures, supervised lab work, and provided academic support for undergraduate and postgraduate courses, including **Computer Vision, Compiler Design, C/C++ Programming and OOP, Design and Analysis of Algorithms, Computer Networks, and Logic Design & Computer Architecture**
- Supported and supervised lab sessions, helping students apply practical concepts and debug code
- Served on the evaluation committee for the **Nile University Undergraduate Research Forum (UGRF)**, assessing student-led research and graduation projects across disciplines
- Mentored students in **research methodology, scientific communication, and project development** to promote a culture of innovation

Reserve Officer

» **Egyptian Armed Forces**

Nov 2017 – Apr 2020 | Egypt

- Established a strong technical foundation through rigorous self-study in **Data Science, Machine Learning, and Deep Learning**, laying the groundwork for advanced research and career transition into AI.
- Developed leadership and team management skills while coordinating operational tasks in a high-discipline environment.

EDUCATION

Master's Degree in Informatics (Artificial Intelligence)

» **ITCS School | Nile University**

September 2021 – Ongoing | GPA: 3.7

- **Thesis:** STREAMTRACK
  A Streams-and-Tables Architecture for Scalable and Reconfigurable Multi-Camera Object Tracking
  Research areas: Multi-Camera Object Tracking, Real-Time Stream Processing, Distributed Systems

  - Conducted research on scalable multi-camera object tracking, developing an event-driven architecture that models detections as streams and maintains global identities and camera topology as queryable relational state.
  - Evaluated the architecture against a functionally equivalent batch baseline, achieving **9.8× end-to-end speedup**, and demonstrated topology-aware association with up to **4.16× lower latency**.
  - **Research Publication:** *STREAMTRACK: A Streams-and-Tables Architecture for Scalable and Reconfigurable Multi-Camera Object Tracking*.
    Accepted at NILES2026; IEEE Xplore publication pending. Related journal manuscript under review at Future Generation Computer Systems.

- Relevant coursework: **Machine Learning, Deep Learning, Computer Vision, Image Forensics, Software Design, and Advanced Big Data**.
- Selected research and engineering projects:
  - YOLOv4_Mosaic-2_NoPainNoGain – Object Detection and Deep Learning.
  - Intelligent-Traffic-Light-Control-Using-Image-processing – Computer Vision and Real-Time Image Processing.
  - G-Research_Crypto_Forecasting – Machine Learning and Time-Series Forecasting.

Bachelor's Degree in Electronics and Electrical Communications Engineering

Faculty of Engineering, Cairo University

September 2012 – July 2017 Graduated with Very Good

- **Graduation Project (Grade: Excellent): Self-Organizing Telecom Network (SON) using Machine Learning.** Designed and implemented an intelligent telecom network management system for *autonomous fault detection, self-healing, and performance optimization* using machine learning
- Developed ML-based models for **network anomaly detection, failure prediction, and real-time corrective action**, with adaptive optimization of network parameters for improved QoS, coverage, and resource utilization

TECHNICAL SKILLS

AI & Machine Learning: PyTorch, TensorFlow, Deep Learning, Model Evaluation, ONNX, CUDA

Programming Languages: Python, C++, C, Java

AI Inference & Model Optimization: TensorRT, ONNX Runtime, NVIDIA Triton Inference Server, NVIDIA DeepStream, Real-Time Inference, GPU Acceleration

AI Pipelines & Distributed Systems: Microservices, NATS, gRPC, REST APIs, Event-Driven Architecture, Streaming Pipelines, Complex Event Processing (Esper), MongoDB

MLOps & ML Lifecycle: Docker, CI/CD, MLflow, Weights & Biases, Model Versioning, Dataset Management, Monitoring & Observability

Computer Vision: OpenCV, YOLO (Ultralytics), RT-DETR, MediaPipe, SAHI, Multi-Camera Video Analytics

Embedded Systems & Edge Computing: ARM Cortex-M, STM32, Nordic nRF, TI Tiva C, NXP, FreeRTOS, QNX, Bare-Metal Programming, Secure Bootloaders, FOTA/OTA, HAL Development

Communication Protocols: BLE, I2C, UART, SPI

Engineering Tools & Systems: Linux, Git, CMake, QEMU, Ceedling, Linux Systems Programming, Logic Analyzers, USB Sniffers, FTDI Tools

LANGUAGES

Arabic — Native language

English — Fluent TOEFL iBT 91